AMRIT PUHAN

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SUMMARY

Interested in how reliable collective judgments can be drawn from noisy, disagreeing people, and how that question carries over to evaluating and aligning AI systems with human preferences.

EDUCATION

Master of Science, Informatics (Data Science Concentration) 08/2022 – 08/2024
Pennsylvania State University, University Park, PA, USA GPA: 3.88/4.0
Thesis: *Recovering Ground Truth Rankings When the Majority Is Misinformed* (**Best MS Thesis Award in AI**)
Advisor: Dr. Hadi Hosseini

Bachelor of Technology, Computer Science and Engineering 07/2016 – 05/2020
National Institute of Technology, Rourkela, India GPA: 3.7/4.0
Thesis: *Human Gait as a Cognitive State Predictor*
Advisor: Dr. Anup Nandy

PUBLICATIONS

*First author; authors listed alphabetically. †Equal contribution, co-first author.

1. Hosseini, H., Mandal, D., **Puhan, A.*** SP-Rank: A Dataset for Ranked Preferences with Secondary Information. *KDD 2026*, Datasets and Benchmarks Track (Poster).
2. Hosseini, H., Mandal, D., **Puhan, A.*** Surprisingly Popular Voting with Concentric Rank-Order Models. *The ACM Web Conference (WWW) 2025*, Main Track (Poster), pp. 3026–3036.
3. Hosseini, H., Mandal, D., **Puhan, A.*** The Surprising Effectiveness of SP Voting with Partial Preferences. *NeurIPS 2024*, Main Track (Poster), pp. 2787–2829.
4. Hazra, S., Pisipati, M., **Puhan, A.**, Nandy, A., Scherer, R. Two Novel Methods for Multiple Kinect v2 Sensor Calibration. *CVIP 2021*, Springer, pp. 403–414.
5. Pisipati, M.†, **Puhan, A.**†, Kumar, A., Semwal, V.B., Agrawal, H. A Dynamic Hand Gesture-Based Password Recognition System. *ICCIS 2019*, Springer, pp. 21–34.

RESEARCH EXPERIENCE

FAIR Lab, Penn State, Graduate Researcher 01/2023 – 12/2024
Advised by Dr. Hadi Hosseini; collaborated with Dr. Debmalya Mandal (University of Warwick)

Recovering Ground Truth Rankings When the Majority Is Misinformed (MS Thesis)

- Designed scalable SP voting algorithms that recover ground-truth rankings from partial preferences when experts are a minority. The best variant (Partial-SP) recovers the true ranking about **4x** more accurately than Copeland on votes alone. *(NeurIPS 2024)*
- Modeled voter behavior as concentric mixtures of Mallows and Plackett-Luce models, inferred parameters with Bayesian MCMC (No-U-Turn Sampler), and derived when SP voting recovers the ground truth with high probability. On real data, SP voting made about **6x** fewer ranking errors than Copeland. *(NeurIPS 2024, WWW 2025)*
- Released **SP-Rank**, the first large-scale public dataset pairing personal votes with predictions of others' votes. It holds **12,384** datapoints from **1,152** participants across 9 elicitation formats, open-sourced with benchmarks showing SP voting recovers rankings about **5x** more accurately than standard Copeland voting. *(KDD 2026)*

LLM fine-tuning and preference learning

- Used SP voting to generate preference labels for RLHF reward-model training, with weak and strong language models as voters, improving label accuracy by **9%** on average over standard voting rules in preliminary experiments. *(MS Thesis)*
- Fine-tuned LLMs for multiple-choice QA with LoRA, quantization and PEFT. Ranked **6th** on the AI2 SWAG leaderboard with **90.98%** accuracy (2024).

Intelligent Computing and Computer Vision Group, NIT Rourkela, Undergraduate Researcher 08/2019 – 05/2020
Advised by Dr. Anup Nandy

- Proposed two calibration methods for an 8-sensor Microsoft Kinect v2 gait-capture setup, reaching a mean reprojection error of **0.42 pixels**, a **76.27%** reduction over existing methods. *(CVIP 2021)*

- Predicted cognitive state from calibrated gait features with ML models ($R^2 = \mathbf{0.9556}$), suggesting gait is a strong indicator of cognitive state.
- Co-developed a hand-gesture password recognition system that runs on an ordinary laptop webcam, reaching **96.16%** test accuracy and outperforming prior webcam (93.75%) and Kinect-based (86.5%) systems. *(ICCIS 2019)*

TEACHING EXPERIENCE

Penn State College of IST, Instructional Assistant, IST 261: Application Development (Java, MVC) Spring 2023

- Coached a 20+ student undergraduate class in object-oriented design, MVC architecture and debugging through office hours and one-on-one assignment help.
- Graded assignments with constructive feedback and served as a liaison between students and faculty.

ACADEMIC SERVICE

Reviewer, Internal Amazon Science Conference 2026
Reviewer, KDD 2026, Datasets and Benchmarks Track
Reviewer, The ACM Web Conference (WWW) 2025, Main Track

WORK EXPERIENCE

Amazon Web Services, Software Development Engineer II, Applied AI Solutions (Seattle, WA) 12/2024 – Present

- Build AI systems over **petabyte-scale** automotive and industrial data at **sub-second latency**.
- Develop a multi-modal video search system for autonomous-driving (ADAS/AV) teams, helping developers surface rare edge cases across video, sensor and annotation data with plain-English queries.
- Develop time-series anomaly detection for predictive maintenance of industrial equipment, evaluating supervised and unsupervised statistical and deep-learning models.

Penn State, Graduate Assistant, Organizational Effectiveness (State College, PA) 08/2023 – 08/2024

- Built Power BI enrollment dashboards for Penn State World Campus and a reusable template that cut development time by **60%** and improved tracked enrollment KPIs by **25%**.

SAP Labs India, Associate Software Developer (Bengaluru, India) 07/2020 – 07/2022

- Developed **five** front-end applications for SAP SuccessFactors (enterprise B2B cloud).
- Quality Analyst for **six** projects; coordinated the product-wide quarterly testing event **twice**.
- CI/CD contact for **ten** teams; automated Jenkins deployments of **40+** applications across **five** countries.

Schneider Electric India, Summer Research Intern (Bengaluru, India) 05/2019 – 07/2019

- Built a cross-platform IoT door-entry prototype (web, Android, iOS) with real-time text, audio and video, cutting latency by **83.3%**.

HONORS AND AWARDS

- **Best MS Thesis Award in Artificial Intelligence**, third annual Penn State AI Awards (2025).
- **NeurIPS 2024 Scholar Award.** Given to select early-career researchers; grant covering registration and travel.
- **Fast Learner Award**, SAP Labs India (2021).
- **Top 1%, JEE Main.** Among 1.4 million candidates (Ministry of Education, Govt. of India).
- **National Talent Search Examination Scholar**, NCERT (2012). National rank 245 of 1M+.
- **Junior Mathematics Olympiad Scholarship**, IMA, Govt. of Odisha (2012). State rank 8.